RR-F-

RR-F-00425A(GSA-FSS)
August 11, 1969
SUPERSEDING
Int. Fed. Spec. RR-F-00425(GSA-FSS)
REVISION I
September 10, 1956

INTERIM FEDERAL SPECIFICATION

FLARES, MARKER, GROUND OBSTRUCTION (LIQUID-BURNING, EMERGENCY)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon current available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The ground obstruction marker flares covered by this specification, display a warning light, indicating to the driver of an approaching vehicle that beyond the burning flare there is a hazard and he should proceed with caution. The fuel used in the flare is kerosene or other liquid fuel having a fire hazard classification not greater than kerosene.

1.2 Classification.

1.2.1 Types and classes. The flares covered by this specification shall be of the following types, classes, styles, and duties, as specified (see 6.1):

Type I - Automotive type (set of three).
  Class A - Flat bottom, nonself-righting.
    Style 1 - In metal box (fig. 1).
      12-hour duty - Minimum of 12 hours burning time per flare.
    Style 2 - In metal racks (fig. 2).
      12-hour duty - Minimum of 12 hours burning time per flare.

Type II - Highway-construction type.
  Class A - Flat bottom, nonself-righting (fig. 3).
    24-hour duty - Minimum of 24 hours burning time per flare.
    36-hour duty - Minimum of 36 hours burning time per flare.
  Class B - Round bottom, self-righting (fig. 4).
    24-hour duty - Minimum of 24 hours burning time per flare.
    36-hour duty - Minimum of 36 hours burning time per flare.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

PPP-B-636	- Box, Fiberboard.
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shape.

Federal Standards:

Fed. Std. No. 123	- Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141	- Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

FSC 6260

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM):

B 117-64 - Salt Spray (Fog) Testing.

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Society of Automotive Engineers, Inc., (SAE):

J 597 - Standard for Liquid Burning Emergency Flares.

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, New York 10017.)

Underwriters' Laboratories (UL), Inc. Standards:

UL 912 - Highway Emergency Signals - Liquid-Burning Flares.

(Copies may be obtained from the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, New York 11749; 207 East Ohio Street, Chicago, Ill. 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

National Classification Board:

National Motor Freight Classification.

(Applications for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown are for the convenience of identification and are not intended to preclude the purchase of flares which are otherwise in accordance with the requirements of this specification (see 4.4.1.2). Type I flares as illustrated in figure 1 are shown in a box in a vertical position. Flares carried in a box in a horizontal position are also acceptable provided they are in accordance with the requirements of this specification.

3.2 Design and construction. The flares shall be designed for use with kerosene or other liquid fuel having a fire hazard not greater than kerosene. Joints, if any, shall be constructed in a manner to prevent leakage. The burner element shall be securely fastened in its proper position. The flares when lighted shall be discernible at a distance of 500 feet under normal atmospheric conditions and low visibility. All 12-hour duty flares covered by this specification shall meet the applicalbe test requirements of section 4, which are essentially the same as the test requirements for liquid burning emergency flares in the Society of Automotive Engineers' Handbook SAE J597 and Underwriters' Laboratories, Inc., Standard UL 912; the longer duty flares shall meet the longer minimum burning time requirements as outlined in section 4.

3.3 Marking. Flares shall be marked in a plain and permanent manner with the manufacturer's name or trade-mark of such known character that the source of manufacture may be readily determined.

3.4 Finish. Unless otherwise specified, the color of the flares shall be the manufacturer's standard (see 6.1). The finish shall withstand the corrosion test requirements of section 4.

3.5 Duty. Each flare covered by this specification, when filled with kerosene and the wick lighted, shall burn at least the minimum number of hours specified, when tested in accordance with section 4.

3.6 Type I. Type I flares shall be furnished in sets of three, mounted in either a metal box (see 3.6.2.1) or metal rack (see 3.6.2.2) dependent upon the style specified. The metal box or rack shall be provided with a means whereby it can be locked although the lock need not be furnished. When specified, each type I set of 3 flares shall be furnished with a set of 3 red flags and 3 flagstaffs (see fig. 1 and 6.1). Each type I flare shall be furnished with a wick at least 9 inches long. Type I flares shall be provided with a cap unless it is so designed that stacking (nesting) will extenguish the flare and prevent leakage of fuel when not in use. The metal thickness for type I flares shall be not less than 0.029 inch.

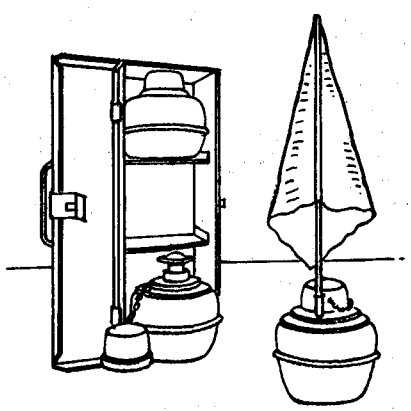


Figure 1. Type I, class A, style 1 flares, automotive type (set of 3), flat bottom, nonself-righting, in metal box. (see 3.1 re vertical vs. horizontal arrangement of flares in box.)

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3.6.1 Flags. The flags when furnished shall be at least 12 x 12 inches square. They shall be of fast color red cotton cloth, the edges of which shall be hemmed or otherwise finished to prevent raveling. The flagstaff shall be either of wood or metal. Each flagstaff shall be at least 15 inches in length and shall fit snugly into the socket on the flare.

3.6.2 Class A. Type I, class A flares shall have a flat bottom that is not weighted. For the purpose of this specification, a flat bottom flare shall include any flare having either a completely flat bottom or a bottom which rests on a flat surface on the outer portion and is recessed or otherwise concave to permit nesting or closer mounting of the flares.

3.6.2.1 Style 1. Each set of three, type I, class A, style 1 flares shall be furnished in a substantial metal box suitable for mounting on a truck. When flags are specified, the metal box shall be of such size as to retain the 3 flares, 3 flags, and 3 flagstaffs.

3.6.2.2 Style 2. Each set of three type I class A, style 2 flares shall be furnished on a substantial metal rack, suitable for mounting on a truck (see figure 2). The rack shall provide for ease in mounting and removing flares. When flags are specified, a metal container for attachment to the rack or for independent mounting shall be furnished for carrying the 3 flags and 3 flagstaffs.

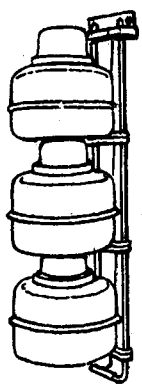


Figure 2. Type I, class A, style 2 flares, automotive type (set of 3), flat bottom, nonself-righting, on metal rack.

3.7 Type II. Type II flares shall be suitable for highway construction and other use. The body of the flare shall be made of sheet steel not less than 0.035 inch in wall thickness. Each type II, 36-hour duty flare shall be furnished with a wick at least 3/4-inch in diameter and at least 18 inches long. Each type II, 24-hour duty flare shall be furnished with a wick at least 9 inches long. When specified, type II flares shall be provided with a cap unless it is so designed that stacking (nesting) will extinguish the flame and prevent leakage of fuel when not in use (see 6.1).

3.7.1 Class A. Type II, class A, flat bottom flares shall conform to 3.6.2 (see figure 3).

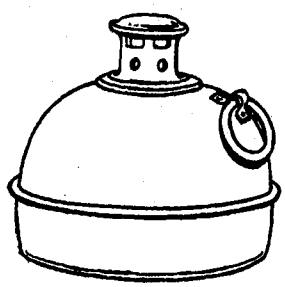


Figure 3. Type II, class A flare, highway construction type, flat bottom, nonself-righting.

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3.7.2 Class B. Type II, class B, flares shall be provided with a weighted and sufficiently rounded or spherical bottom to return the flare to an upright position in the event it is tipped over (see figure 4).

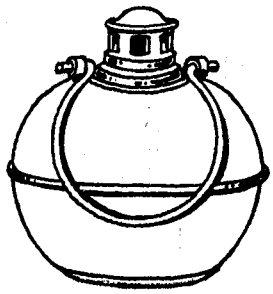


Figure 4. Type II, class B flare, highway construction type, round bottom, self-righting.

3.8 Instructions for use. The manufacturer shall furnish with each set of flares printed instructions as to wick adjustment, maximum filling level, the method of installation, and such other details as the manufacturer may deem necessary in order that the user of these flares may be able to operate them at their maximum efficiency.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Visual inspection. Sampling for visual inspection shall be conducted in accordance with MIL-STD-105, inspection level S2, acceptable quality level (AQL) 4.0 percent defective.

4.2.2 Tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S3, AQL 2.5.

4.3 Inspection. A visual inspection shall be conducted in accordance with, but not limited to, the following:

Materials, parts, or accessories missing or not as specified.
Finish not as specified.

4.4 Tests.

4.4.1 Vibration and shock test. Three type I sample flares, in the metal rack or metal box in which they were submitted to the laboratory, positioned as would be normal in service, shall be bolted to the anvil end of the table of the vibration-test machine (figure 5) and vibrated approximately 750 times per minute through a distance of 1/4-inch. The table shall be spring-mounted at one end and fitted with steel calks on the underside of the end. These calks are to make contact with the steel anvil once during each cycle at the completion of the fall. The rack shall be operated under a spring tension of 60 to 70 pounds. This test shall be continued for one hour.

4.4.1.1 At completion of the test specified in 4.4.1, the flares shall be examined. Any flare showing evidence of structural failure or leakage in the body or joints at any time during this test fails to meet the requirements of this specification.

4.4.1.2 It is recommended that for the purpose of standardizing the vibration and shock test, it be made on a testing machine substantially in accordance with figure 5.

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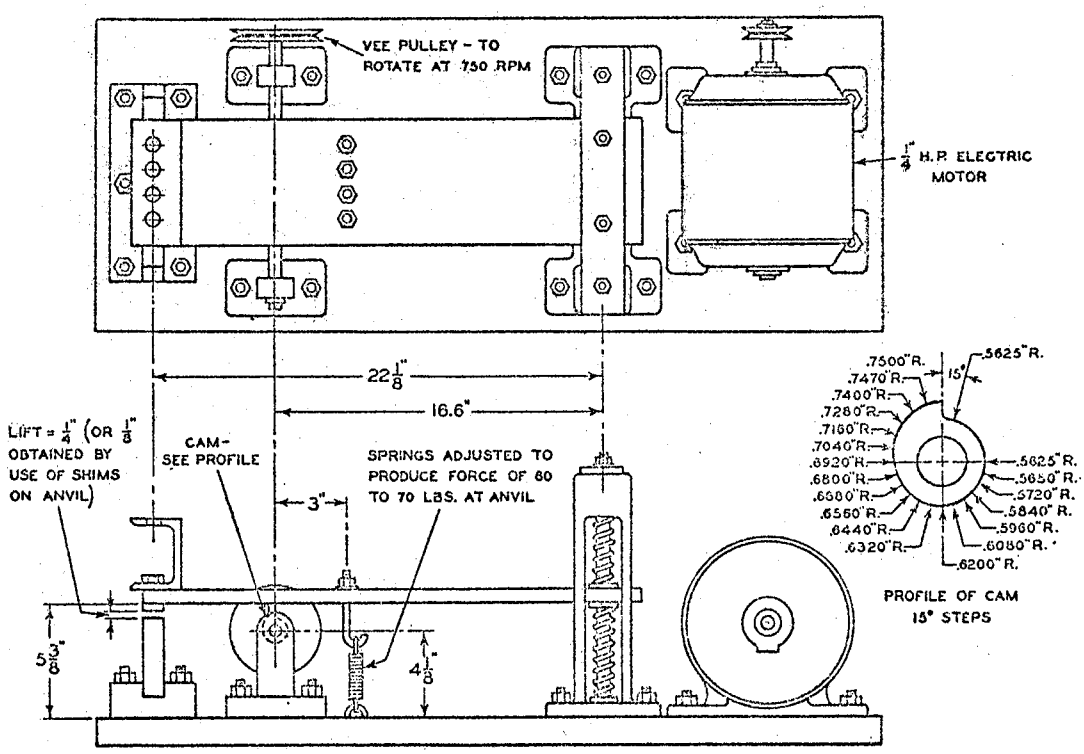


Figure 5. Vibration and shock-testing machine.

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4.4.2 Corrosion test. A sample unit shall be subjected to a salt spray (fog) test in accordance with Fed. Test Method Std. No. 141, method 6061 or ASTM B-117-64 for a period of 50 hours consisting of two periods of 24 hours exposure and one-hour minimum drying time each. There shall be no evidence of excessive corrosion, immediately after the above test has been completed, which would affect the proper functioning of the flare.

4.4.3 Weatherproof test. The flares used for these tests shall be filled with kerosene to the level recommended in the manufacturer's instructions and shall have the wicks adjusted as recommended by the manufacturer.

4.4.3.1 After a preheating period of 5 minutes in still air, a sample flare, lighted and mounted in its normal operating position on a table rotating at 4 revolutions per minute and in a wind of approximately 2 miles per hour, shall be subjected to a water spray from an adjustable, solid-cone nozzle (such as the ordinary garden-hose spray nozzle), set so that the nozzle outlet is 8 to 12 feet horizontally from the sample and 1 to 3 feet vertically above the sample, with the nozzle axis pointing upward at an angle of approximately 45 degrees from the horizontal and with the water striking the sample at an angle of approximately 45 degrees from the horizontal in a downward direction.

4.4.3.2 Under the conditions specified above and with a water pressure of 5 to 7-1/2 pounds per square inch at the nozzle, the rate of precipitation at the location of the sample shall be adjusted to 0.10 inch per minute. The sample flare shall be introduced gradually into the spray and after being placed in the test location shall continue to operate under these conditions for 15 minutes. This test shall be made on each of the three flares constituting a set, and at least two out of three flares shall pass the test.

4.4.3.3 With the rate of rotation and the wind condition the same as described above, the water pressure shall be further increased to a value of 18 to 20 p.s.i. and the sample moved farther from the nozzle, if necessary, to a location giving a precipitation of 0.01 inch per minute, striking the sample at approximately 45 degrees from the horizontal. Under these conditions, the sample shall continue to operate for 1 hour and 30 minutes. This test shall be made on each of the three flares constituting a set, and at least two out of three flares shall pass the test.

4.4.4 Reliability, stability, and life tests. The flares used for these tests shall be filled with kerosene to the level recommended by the manufacturer and shall have the wicks adjusted in accordance with the manufacturer's instructions.

4.4.4.1 A flare that is nearly empty of fuel shall be lighted, placed on a horizontal concrete surface similar to a pavement, and allowed to burn in still air for a preheating period of 5 minutes. It shall then be placed suddenly in an airstream of 40 miles per hour and withdrawn. After rotating through approximately 45 degrees, it shall again be placed in the air stream and withdrawn. It shall be rotated again through an additional 45 degrees approximately and placed in the air stream a third time. If the flame is extinguished or if the flare slides or is overturned by the wind on any one of these three operations, the sample shall have failed to pass the test. The test shall then be repeated on each of the other two samples constituting a set and at least two out of three flares shall pass the test.

4.4.4.2 A flare, lighted and mounted in its normal operating position, rotating about its vertical axis at 4 revolutions per minute, shall be subjected to a horizontal current of air having a velocity the equivalent of wind at 40 m.p.h. This test shall continue for 15 minutes and the flare shall remain lighted throughout the entire 15-minute period.

4.4.4.3 Upon completion of the above test, the flare shall, while lighted and rotated as specified above, be subjected to a horizontal current of air having a velocity of 5 m.p.h. The total uninterrupted burning time, including the first 15 minutes at 40 m.p.h., shall be at least 12 hours. The flare shall not be refilled during tests; there shall be no adjustment of wick.

4.4.4.4 The flare shall operate satisfactorily for a period of at least 2 hours in still air.

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4.4.4.5 Photometric test. A sample flare, when subjected to a wind velocity of 5 m.p.h. and 40 m.p.h., respectively, shall produce a minimum of 0.10 candle power in a horizontal direction.

4.4.5 Type II flare tests. Type II flares shall be subjected to the same test requirements as type I flares except for the substitution of the drop test for the vibration and shock test and the longer minimum burning time test (see 4.4.5.1 and 4.4.5.2).

4.4.5.1 Drop test. In lieu of the vibration and shock test (4.4.1 to 4.4.1.2 inclusive) for type I flares, the type II flares shall be subjected to a drop test. Each empty flare shall be dropped on a concrete pavement 4 times from a height of 4 feet. After the drop test, each sample flare shall be refilled with kerosene and allowed to stand at least one hour, the flares shall then be examined and shall show no leakage. This test shall be made on at least four flares and all four flares shall pass the test.

4.4.5.2 Uninterrupted burning time test. In lieu of the minimum uninterrupted burning time of 12 hours as required of type I flares (see 4.4.4.3), the type II flares when tested under the same conditions as type I flares, shall have a minimum uninterrupted burning time as the duty specified, the 24-hour duty flares shall burn at least 24 hours and the 36-hour duty flares shall burn at least 36 hours.

4.5 Examination for preparation for delivery. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored as specified in table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

Table I. Examination for preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	More or less than required.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper, or missing.

4.6 Examination for closure and strapping of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging is not required.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Flares in quantities as specified (see 6.1), shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, class weather resistant. Closure shall be in accordance with the appendix of that specification.

5.2.2 Level B. Twelve marker flares shall be packed in a fiberboard box conforming to PPP-B-636, class domestic. The box shall be fitted with half-slotted partitions as high as the interior of the box. The partitions shall be made of fiberboard conforming to PPP-F-320, class domestic to provide a separate cell for each flare.

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5.2.3 Level C. Flares in quantities as specified (see 6.1), shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking of the exterior shipping container shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, marking of the exterior shipping container shall be in accordance with MIL-STD-129.

6. Notes:

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and style desired (see 1.2).
(c) Color of flare, if other than specified (see 3.4).
(d) If red flags are required for type I flares, so state (see 3.6).
(e) If cap for extinguishing flame is required, so state (see 3.7).
(f) Responsibility for inspection, if other than specified (see 4.1).
(g) Level of packing required (see 5.2).
(h) Quantities required (see 5.2.3).
(i) Marking, if other than specified (see 3.3 and 5.3).

Preparing activity:

GSA - FSS

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

RR-F-00425A(GSA-FSS) - Flares, Marker, Ground Obstruction (Liquid-Burning, Emergency)

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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